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C5 — Monitoring and Controlling Chemical Reactions

OCR GCSE Chemistry A (J248) — Gateway Science · Papers 2 & 4

C5.1 Monitoring Chemical Reactions

> Chemists **monitor** reactions by measuring amounts: the **concentration** of solutions, the **volume** of gases and the **mass** of products. All of these link back to the **mole** — the unit of *amount of substance* (not mass, not a number of particles).

Before you start, remember the key mole equation from C3: **amount (mol) = mass (g) ÷ M_r**. Almost every calculation in this topic is a chain of steps that goes *through moles*.

Concentration in mol/dm³ (HT) — C5.1a

> The **concentration** of a solution is the amount of solute dissolved in a given volume of solution. In mol/dm³ it is the number of moles of solute in **1 dm³** (1000 cm³) of solution.

concentration (mol/dm³) = amount of solute (mol) ÷ volume of solution (dm³)

and amount (mol) = mass of solute (g) ÷ M_r · volume in dm³ = volume in cm³ ÷ 1000

So the concentration depends on **both** the mass of solute and the volume of solution: dissolving more solute in the same volume increases the concentration; dissolving the same solute in a larger volume decreases it.

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Worked Example: 4.0 g of sodium hydroxide (NaOH, M_r = 40) is dissolved in water and made up to 250 cm³. Find the concentration in mol/dm³.
amount = 4.0 ÷ 40 = 0.100 mol · volume = 250 ÷ 1000 = 0.250 dm³
concentration = 0.100 ÷ 0.250 = **0.400 mol/dm³**

Making a standard solution (a solution of accurately known concentration): weigh the solid accurately on a balance, dissolve it in a little distilled water in a beaker, transfer the solution *and all the washings* into a volumetric flask, make up to the graduation mark with distilled water (bottom of the meniscus on the line) and invert several times to mix.

Concentration in g/dm³ (HT) — C5.1f

concentration (g/dm³) = mass of solute (g) ÷ volume of solution (dm³)

To convert: concentration (g/dm³) = concentration (mol/dm³) × M_r

Worked Example: The 0.400 mol/dm³ NaOH solution above: 4.0 g ÷ 0.250 dm³ = **16 g/dm³** (check: 0.400 × 40 = 16 g/dm³).
What mass of NaOH is in 50.0 cm³ of it? mass = 16 × 0.0500 = **0.80 g**.

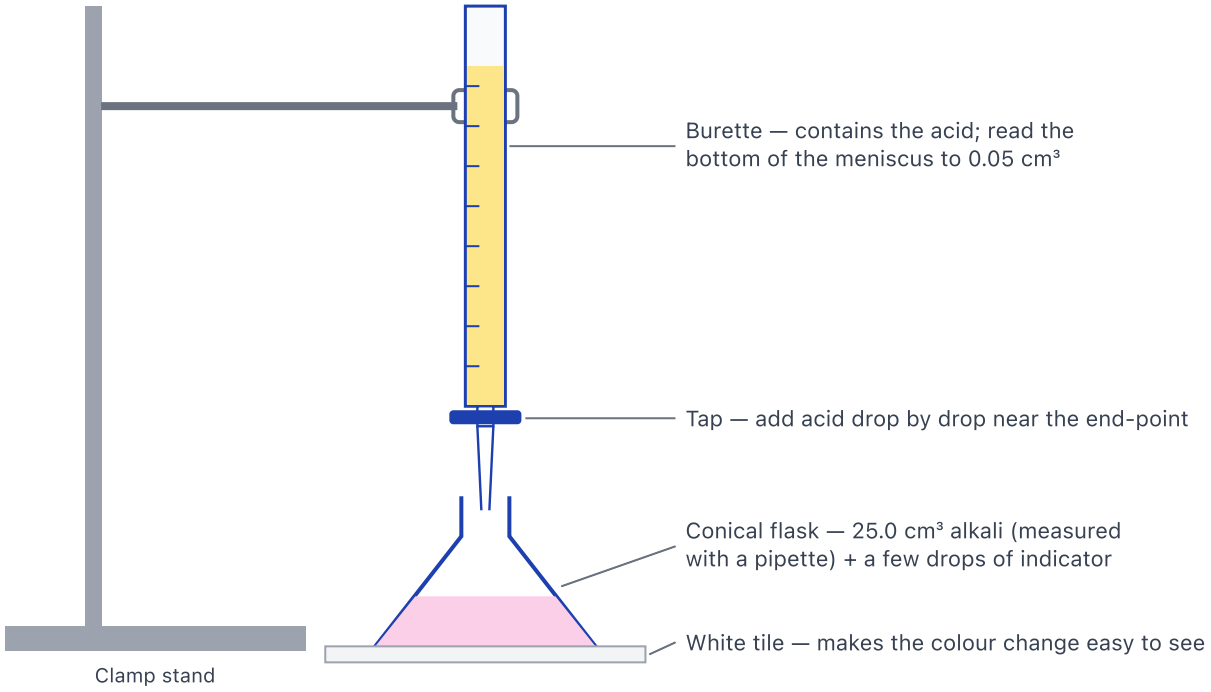
Exam Tip: Rearrange rather than memorise: mass = concentration (g/dm³) × volume (dm³). The most common error is forgetting to divide cm³ by 1000.

Titration — C5.1b

A **titration** is used to find the exact volume of one solution that reacts completely with a known volume of another — for example, the volume of acid that exactly neutralises 25.0 cm³ of alkali. The point at which the reaction is just complete is the **end-point**, shown by a colour change of an **indicator**.

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Titration Apparatus



1. Use a **pipette** (with a pipette filler) to transfer exactly 25.0 cm³ of alkali into a conical flask. Add a few drops of indicator (e.g. phenolphthalein or methyl orange).
2. Fill the **burette** with acid (using a funnel, then remove it). Make sure the jet below the tap is full. Record the initial reading.
3. Add acid while **swirling** the flask. Near the end-point add it **drop by drop** until the indicator *just* changes colour permanently. Record the final reading.
4. Titre = final reading – initial reading. The first run is a quick **rough** titration.
5. Repeat until you get **concordant** titres (within 0.10 cm³ of each other) and take the mean of these.

Run	Rough	1	2	3
Final reading / cm³	24.60	48.75	24.25	48.45
Initial reading / cm³	0.00	24.60	0.00	24.25
Titre / cm³	24.60	24.15	24.25	24.20

Illustrative data. Concordant titres 24.15, 24.25, 24.20 → mean titre = **24.20 cm³** (the rough titre is ignored).

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Indicator	Colour in acid	Colour in alkali
Phenolphthalein	Colourless	Pink
Methyl orange	Red	Yellow
Litmus	Red	Blue (not used for titrations — colour change too gradual)

Exam Tip: A **single-colour-change** indicator is used (not universal indicator) because you need a sharp end-point. A pH probe/meter can also be used to follow a neutralisation (PAG C6).

Titration Calculations (HT) — C5.1c

The volume and concentration of one solution give you its **amount in moles**. The balanced equation gives the **mole ratio**. That ratio gives the moles of the other substance, from which you can find its concentration or volume.

$$\text{amount (mol)} = \text{concentration (mol/dm}^3\text{)} \times \text{volume (dm}^3\text{)}$$

Worked Example: 25.0 cm³ of 0.100 mol/dm³ NaOH is neutralised by 20.0 cm³ of HCl.
$NaOH + HCl \rightarrow NaCl + H_2O$
1. moles NaOH = 0.100 × (25.0 ÷ 1000) = 0.00250 mol
2. ratio 1 : 1, so moles HCl = 0.00250 mol
3. concentration HCl = 0.00250 ÷ (20.0 ÷ 1000) = **0.125 mol/dm³**

Worked Example: 25.0 cm³ of 0.200 mol/dm³ NaOH reacts with 18.75 cm³ of sulfuric acid.
$H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$
moles NaOH = 0.200 × 0.0250 = 0.00500 mol → moles H_2SO_4 = 0.00500 ÷ 2 = 0.00250 mol
concentration H_2SO_4 = 0.00250 ÷ 0.01875 = **0.133 mol/dm³** (3 s.f.)

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Exam Tip: Always use the mole ratio from the *balanced* equation — diprotic acids such as H_2SO_4 need half as many moles as the NaOH. Give the answer to the same number of significant figures as the data (usually 3).

Moles and Gas Volumes (HT) — C5.1d, C5.1e

At the same temperature and pressure, **equal amounts (moles) of any gas occupy the same volume**. At room temperature and pressure (RTP) one mole of any gas occupies **24 dm³** (24 000 cm³) — the **molar gas volume**.

volume of gas (dm³) = amount (mol) × 24 amount (mol) = volume (dm³) ÷ 24 (at RTP)

This means that for reacting gases the **volume ratio equals the mole ratio** in the balanced equation. In $N_2 + 3H_2 \rightarrow 2NH_3$, 30 dm³ of hydrogen reacts with 10 dm³ of nitrogen to make 20 dm³ of ammonia.

Worked Example: What volume of CO_2 (at RTP) is made when 5.00 g of calcium carbonate reacts with excess acid?

$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$ (M_r $CaCO_3$ = 100)

moles $CaCO_3$ = 5.00 ÷ 100 = 0.0500 mol → moles CO_2 = 0.0500 mol (1 : 1)

volume = 0.0500 × 24 = **1.20 dm³ (1200 cm³)**

In the lab you measure gas volumes with a **gas syringe** or by collecting over water in an inverted measuring cylinder (PAG C8). Dividing the measured volume by 24 000 cm³ gives the amount in moles.

Theoretical Mass of Product — C5.1g

The balanced equation tells you how many moles of product form from a given amount of reactant. Three steps: **mass → moles → use ratio → mass**.

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Worked Example: What mass of magnesium oxide forms when 6.0 g of magnesium burns completely? $2Mg + O_2 \rightarrow 2MgO$ (A_r Mg = 24, M_r MgO = 40)

moles Mg = 6.0 ÷ 24 = 0.25 mol → ratio 2 : 2 → moles MgO = 0.25 mol

mass MgO = 0.25 × 40 = **10 g** (this is the *theoretical* maximum mass)

Percentage Yield — C5.1h

percentage yield = (actual mass of product ÷ theoretical maximum mass) × 100

Worked Example: In the experiment above only 8.0 g of MgO was collected. % yield = (8.0 ÷ 10) × 100 = **80%**.

Yields are less than 100% because: the reaction may be **reversible** and not go to completion; some product is **lost** when it is separated (filtering, transferring, evaporating); some reactants may undergo **side reactions** giving unwanted products; the reactants may be impure.

Atom Economy — C5.1i, C5.1j

The **atom economy** of a reaction is a measure of the proportion of the atoms in the reactants that end up in the **desired product** (rather than in waste by-products).

atom economy = (M_r of desired product ÷ sum of M_r of all reactants) × 100

Use the balanced equation — multiply each M_r by its coefficient.

Worked Example: Ethanol by fermentation: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ (M_r: glucose 180, ethanol 46)

atom economy = (2 × 46) ÷ 180 × 100 = 92 ÷ 180 × 100 = **51.1%**

Ethanol by hydration of ethene: $C_2H_4 + H_2O \rightarrow C_2H_5OH$ atom economy = 46 ÷ (28 + 18) × 100 = **100%** (only one product).

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> **Exam Tip:** Atom economy and percentage yield are different. Atom economy is calculated from the *equation* (theory); percentage yield comes from the *experiment*. A reaction can have 100% atom economy but a low yield.

Choosing a Reaction Pathway (HT) — C5.1k

Industry chooses the route to a product by weighing up several factors, not just one:

- **Atom economy** — high means less waste and better use of resources.
- **Percentage yield** — high means more product from the raw materials.
- **Rate** — a fast reaction gives more product per hour.
- **Equilibrium position** — for reversible reactions, how far the reaction goes under the chosen conditions.
- **Usefulness of by-products** — a by-product that can be sold or used raises the effective atom economy.
- Cost and availability of raw materials, energy needed, and safety/environmental impact.

Factor	Fermentation of sugar	Hydration of ethene
Atom economy	51% (CO_2 by-product)	100%
Rate	Slow (days)	Fast, continuous process
Conditions	~35 °C, yeast, no air	~300 °C, 60–70 atm, phosphoric acid catalyst
Product	Dilute ethanol — needs distilling	Pure ethanol (unreacted ethene recycled)
Raw material	Renewable (crops)	Ethene from crude oil — finite

> **Exam Tip:** In "explain why route X is chosen" questions, use the data given and make a *comparison* for each factor (e.g. "route A has a higher atom economy so less waste is

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produced"), then give an overall judgement.

C5.2 Controlling Reactions

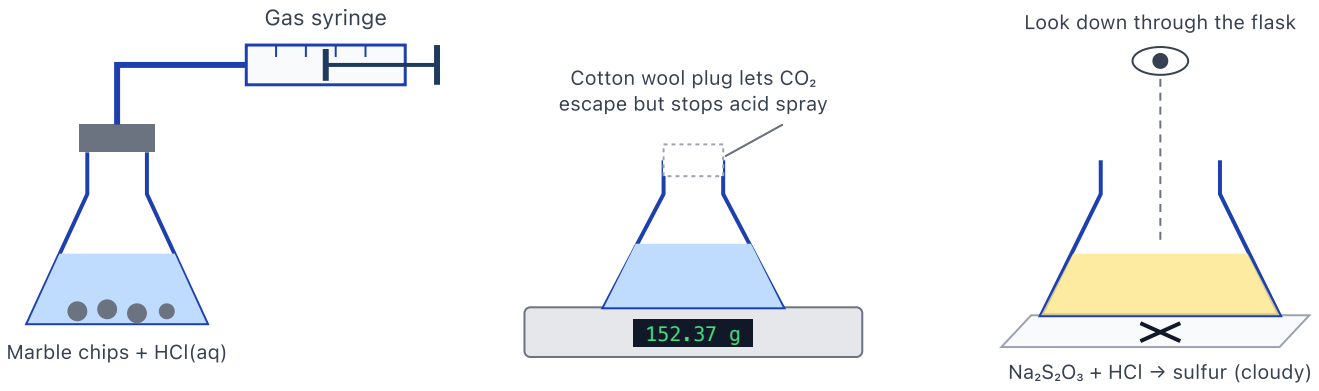
The **rate of reaction** is how quickly reactants are used up or products are formed.

mean rate of reaction = quantity of reactant used (or product formed) ÷ time taken

Units: g/s (mass) or cm³/s (gas volume) · If a reaction takes time t to reach a fixed point, **rate ∝ 1/t**

Practical Methods for Measuring Rate — C5.2a

Three Ways to Follow the Rate of a Reaction



(a) Volume of gas vs time **(b) Loss in mass vs time** **(c) Time for cross to vanish**

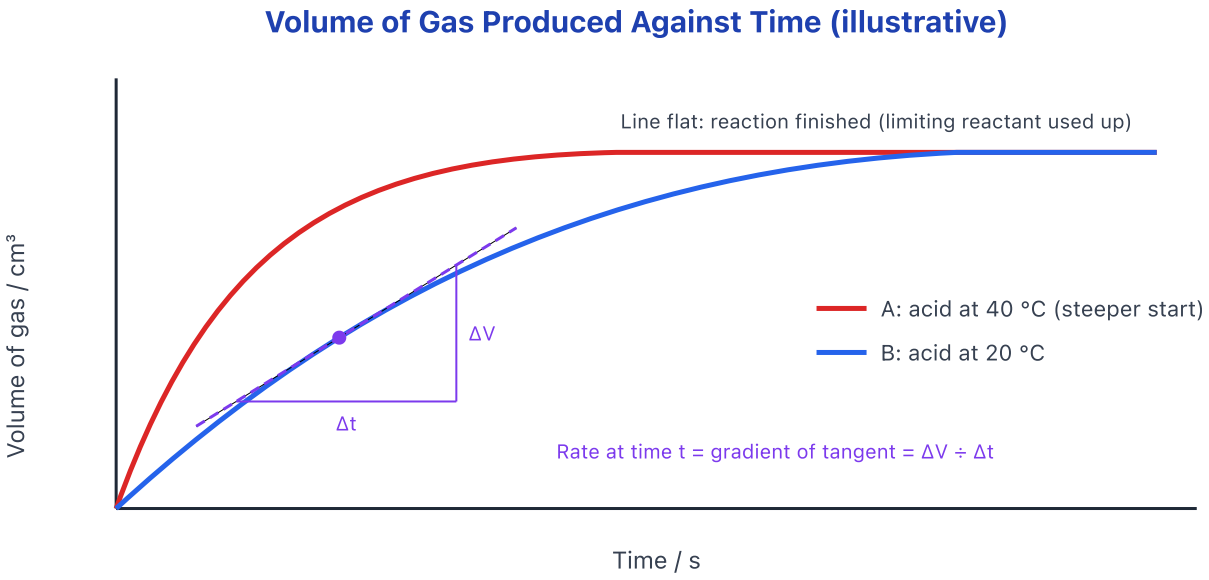
Method	Example reaction	What you measure
Gas collection (gas syringe or upturned measuring cylinder over water)	Magnesium or marble chips + hydrochloric acid	Volume of gas at regular time intervals
Loss in mass (open flask on a balance)	Marble chips + acid (CO_2 escapes)	Mass at regular intervals — falls as gas leaves

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Colour change / precipitate	Sodium thiosulfate + hydrochloric acid ("disappearing cross")	Time for a cross under the flask to disappear
Time for a solid to disappear	Magnesium ribbon in acid	Time for the magnesium to react completely

Keep all other variables constant (control variables) — e.g. same volume and concentration of acid, same mass of solid, same temperature — so it is a fair test. Repeat readings and calculate a mean.

Interpreting Rate Graphs — C5.2b



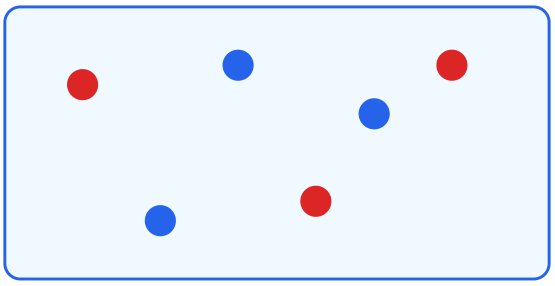
- The **steeper** the line, the **faster** the rate. The gradient is greatest at the start, when reactant concentration is highest.
- The line levels off when one reactant (the **limiting** reactant) is used up. If the same amounts of reactants are used, the final volume is the same — a faster reaction just gets there sooner.
- **Mean rate** over a period = change in volume ÷ time taken (e.g. 60 cm³ in 30 s = 2.0 cm³/s).
- The rate at a particular instant = the **gradient of the tangent** to the curve at that time.
- For "time to reach a fixed point" experiments (disappearing cross), plot **1/t** as a measure of rate: a cross that disappears in 40 s gives 1/t = 0.025 s^{-1}.

Factors Affecting Rate — C5.2c, C5.2d

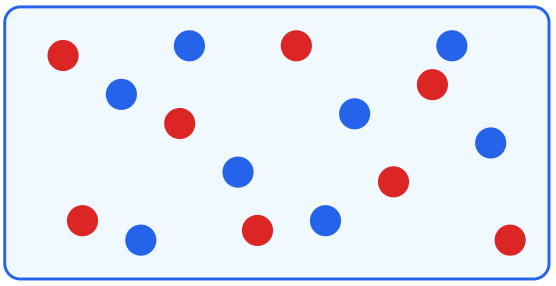
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Collision theory: particles react only when they **collide** with each other with at least a minimum amount of energy — the **activation energy**. The rate depends on the **frequency** of collisions and the **energy** of the collisions (the proportion that are successful).

Higher Concentration → More Frequent Collisions



Low concentration: few particles in the same volume



High concentration: more particles, collide more often

Red = acid particles · blue = other reactant particles

Change	Effect on rate	Explanation (collision theory)
Increase temperature	Increases	Particles have more kinetic energy and move faster, so they collide **more frequently** AND a **greater proportion** of collisions have energy ≥ activation energy (more successful collisions).
Increase concentration (solutions)	Increases	More reactant particles in the same volume, so **more frequent collisions**.
Increase pressure (gases)	Increases	Same number of gas particles in a smaller volume, so **more frequent collisions**.
Increase surface area (smaller pieces / powder)	Increases	More particles of the solid are exposed to the other reactant, so **more frequent collisions**.
Add a catalyst	Increases	Provides an alternative pathway with a **lower activation energy**, so more collisions are successful.

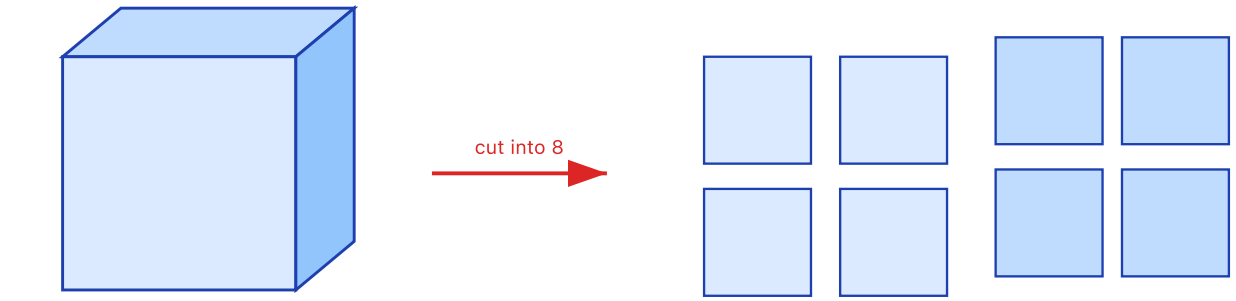
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Exam Tip: Say "more **frequent** collisions" or "more collisions **per second**" — not just "more collisions". Only temperature changes the *energy* of collisions; concentration, pressure and surface area change only the *frequency*.

Surface Area to Volume Ratio — C5.2e

Cutting a solid into smaller pieces does not change its volume, but it increases the total surface area. So the **surface area : volume ratio** increases and more particles are available to collide.

Smaller Pieces Have a Larger Surface Area to Volume Ratio



One 4 cm cube
SA = 96 cm² · V = 64 cm³ · ratio = 1.5

Eight 2 cm cubes
SA = 8 × 24 = 192 cm² · V = 64 cm³ · ratio = 3.0

Worked Example: Same mass of marble: large chips vs powder in excess acid. The powder has the larger surface area : volume ratio, so the reaction is faster (steeper graph) — but the same total volume of CO_2 is produced because the same mass of $CaCO_3$ reacts.

Catalysts — C5.2f, C5.2g, C5.2h

A **catalyst** is a substance that **increases the rate** of a reaction **without being used up** (it is chemically unchanged and has the same mass at the end). Only a small amount is needed, and it can be reused.

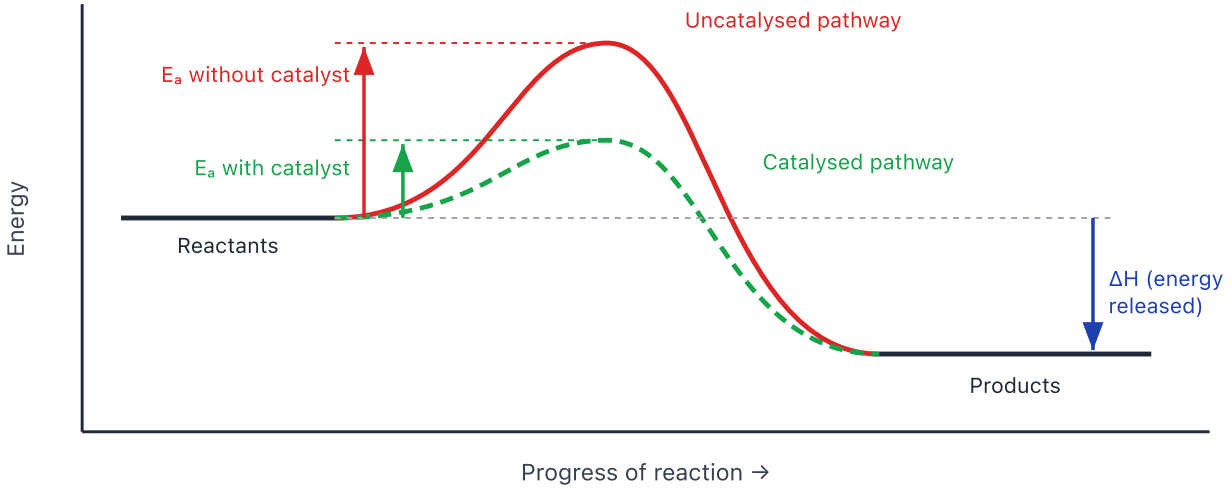
- A catalyst works by providing an **alternative reaction pathway with a lower activation energy**, so a greater proportion of collisions are successful.
- Catalysts are **specific** — a catalyst for one reaction will not necessarily work for another.

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- A catalyst does **not** change the amount of product formed, or the energy change (ΔH) of the reaction.
- **Identifying a catalyst:** it is written above the arrow (or appears on both sides of the overall equation) and is not in the net change; its mass is the same before and after.

Reaction	Catalyst
Decomposition of hydrogen peroxide: $2H_2O_2 \rightarrow 2H_2O + O_2$	Manganese(IV) oxide, MnO_2
Zinc + sulfuric acid → zinc sulfate + hydrogen	Copper (powder or copper(II) sulfate solution)
Haber process: $N_2 + 3H_2 \rightleftharpoons 2NH_3$	Iron
Contact process: $2SO_2 + O_2 \rightleftharpoons 2SO_3$	Vanadium(V) oxide, V_2O_5
Cracking long-chain alkanes	Zeolites / aluminium oxide

Reaction Profile for an Exothermic Reaction, With and Without a Catalyst



Exam Tip: On a reaction profile the catalyst lowers the **peak** only. The reactant and product levels — and therefore ΔH — stay the same. Don't say a catalyst "gives particles more energy".

Enzymes — C5.2i

Enzymes are biological catalysts (proteins) that speed up reactions in living organisms, e.g. amylase breaks down starch, and yeast enzymes catalyse fermentation. They work at mild temperatures, are very specific, and stop working (denature) if too hot or at the wrong pH. Industry uses enzymes to save energy (e.g. biological washing powders, food processing).

C5.3 Equilibria

Reversible Reactions — C5.3a

A **reversible reaction** is one where the products can react to re-form the reactants. The direction can be changed by altering the conditions. The sign ⇌ is used.

$$\text{hydrated copper(II) sulfate (blue)} \rightleftharpoons \text{anhydrous copper(II) sulfate (white)} + \text{water}$$

$$CuSO_4 \cdot 5H_2O \rightleftharpoons CuSO_4 + 5H_2O \quad \text{(heat} \rightarrow \text{forward; add water} \rightarrow \text{reverse)}$$

$$\text{ammonium chloride} \rightleftharpoons \text{ammonia} + \text{hydrogen chloride} \quad NH_4Cl \rightleftharpoons NH_3 + HCl$$

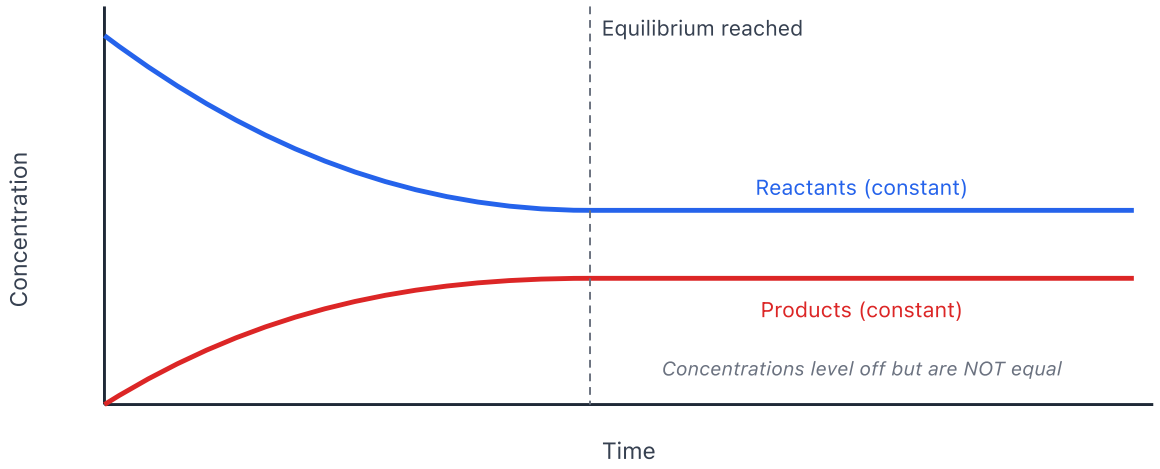
If a reversible reaction is endothermic in one direction it is exothermic by the same amount in the other. Adding water to white anhydrous copper(II) sulfate releases heat.

Dynamic Equilibrium — C5.3b

In a **closed system** (nothing can get in or out), a reversible reaction reaches **dynamic equilibrium** when the **rate of the forward reaction equals the rate of the reverse reaction**. Both reactions are still happening, but the concentrations of reactants and products stay **constant**.

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Approaching Dynamic Equilibrium (illustrative)



> **Exam Tip:** At equilibrium the concentrations are **constant, not equal**, and the reactions have **not stopped** — they continue at equal rates.

Changing the Position of Equilibrium (HT) — C5.3c

> **Le Chatelier's principle:** if a system at equilibrium is subjected to a change in conditions, the position of equilibrium shifts to **oppose the change**.

Example: the Haber process, $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$; the forward reaction is **exothermic**. There are 4 molecules of gas on the left and 2 on the right.

Change	Equilibrium shifts...	Effect on yield of NH_3
Increase concentration of N_2 or H_2	To the right, to use up the added reactant	Increases
Remove NH_3 as it forms	To the right, to replace the product	Increases
Increase **pressure**	Towards the side with **fewer gas molecules** (right: 2 < 4)	Increases
Increase **temperature**	In the **endothermic** direction (left) to take in heat	Decreases

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Decrease temperature	In the **exothermic** direction (right)	Increases (but rate falls)
Add a catalyst	No change — speeds up forward and reverse equally	No change (equilibrium reached sooner)

- If both sides have the **same number of gas molecules**, pressure has no effect on the position.
- To get as much product as possible: choose the temperature that favours the product side (low T if forward is exothermic), high pressure if the product side has fewer gas molecules, and remove product as it forms.
- Colour changes show a shift: e.g. $2NO_2$ (brown) $\rightleftharpoons$ N_2O_4 (colourless), forward exothermic — cooling the mixture makes it paler.

> **Exam Tip:** Always give the *direction* of the shift AND the *reason* (e.g. "shifts right because there are fewer gas molecules on the right, which reduces the pressure"). The compromise conditions used in industry are covered in C6.1.

Practical Activity Groups Linked to C5

PAG	Title	Where it fits in C5
PAG C6	Titration	Acid/alkali titration to find the concentration of an acid (C5.1b, C5.1c); use of pH probes
PAG C8	Measuring rates of reaction	Gas volume vs time, mass loss, disappearing cross, surface area / concentration of an acid–carbonate reaction (C5.1d, C5.2a–g)
PAG C1	Reactivity trend	Magnesium / marble chips with acid; catalysis of H_2O_2 by MnO_2 and of zinc + sulfuric acid by copper (C5.2a–g)
PAG C7	Production of salts	Marble chip / magnesium and acid experiments measuring reaction time or gas volume (C5.2b)

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Quick Summary — Key Equations & Spec Checklist for C5

Quantity	Equation	Units
Concentration (HT)	amount ÷ volume	mol/dm³
Concentration (HT)	mass ÷ volume	g/dm³
Gas volume at RTP (HT)	volume = amount × 24	dm³
Percentage yield	actual ÷ theoretical × 100	%
Atom economy	M_r desired product ÷ total M_r reactants × 100	%
Mean rate	quantity used or formed ÷ time	g/s or cm³/s

Spec	You should be able to...
C5.1a (HT)	Explain how concentration in mol/dm³ relates to mass of solute and volume of solution
C5.1b	Describe the technique of titration (PAG C6)
C5.1c (HT)	Do titration calculations using volumes, concentrations and the mole ratio
C5.1d (HT)	Relate molar amounts of gases to their volumes (equal moles, equal volumes)
C5.1e (HT)	Calculate gas volumes using 24 dm³ per mole at RTP
C5.1f (HT)	Relate mass of solute and volume of solution to concentration (g/dm³)
C5.1g	Calculate the theoretical mass of product from a given mass of reactant
C5.1h	Calculate percentage yield
C5.1i / C5.1j	Define and calculate atom economy from a balanced equation

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C5.1k (HT)	Explain choice of reaction pathway using atom economy, yield, rate, equilibrium and by-products
C5.2a	Suggest practical methods to determine rate (PAG C1, C8)
C5.2b	Interpret rate graphs: gradients, tangents, 1/t
C5.2c / C5.2d	Describe and explain effects of temperature, concentration, pressure, surface area (frequency and energy of collisions)
C5.2e	Explain surface area effects using surface area : volume ratio
C5.2f / C5.2g / C5.2h	Describe catalysts, identify them, explain them using activation energy and reaction profiles
C5.2i	Recall that enzymes are biological catalysts
C5.3a / C5.3b	Recall reversible reactions and dynamic equilibrium in a closed system
C5.3c (HT)	Use Le Chatelier's principle (concentration, temperature, pressure) to predict shifts and choose conditions

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